🧠 Minimal AI Customer Support Agent – Full Stack Challenge

Objective

Build a full-stack application: a simple **AI-powered customer support chat app** that authenticates users, allows them to chat with an AI assistant, and stores the chat history.

🚀 Features to Implement

👤 Authentication (JWT)

- Signup / Login / Logout using JWT
- Passwords hashed using `bcrypt`
- Protected routes

💬 Chat System

- Text input box for user
- Display AI-generated responses
- Save chat history in MongoDB (user-scoped)
- Show past conversations

🧠 AI Integration (Choose One)

- **OpenRouter.ai** (supports many open LLMs with free access)
- **Hugging Face Inference API** (e.g. for DialoGPT or similar)
- Self-hosted or free-tier-compatible models (optional)

📡 Backend (Node.js + Express)

- `/auth/signup`, `/auth/login`
- `/chat/send` (accepts message, returns response)
- `/chat/history` (returns user's chat logs)

🌐 Frontend (React)

- React app with a chat UI
- Use **Axios/Fetch** for API calls
- Handle auth token securely
- Use basic CSS or Tailwind (optional)

🌐 DevOps

- Dockerize the app
- App should work with docker compose up command

📦 Tech Stack

Layer	Tech
Frontend	React (Vite or CRA)
Backend	Node.js + Express
Database	MongoDB Atlas (Free tier)
Auth	JWT + Bcrypt
AI	OpenRouter / Hugging Face
Hosting	Vercel / Netlify / Render / Railway (any free-tier)

⚙️ Bonus (Optional)

- Add typing indicator
- Add rate limiting (`express-rate-limit`)
- Add environment switching (dev/prod configs)

📁 Deliverables

1. GitHub repository with:
    - `client/` and `server/` folders
    - `.env.example` files
    - Clear README with setup steps
2. Deployed link

⏱️ Estimated Time: 6–10 hours (Depending on bonus features)